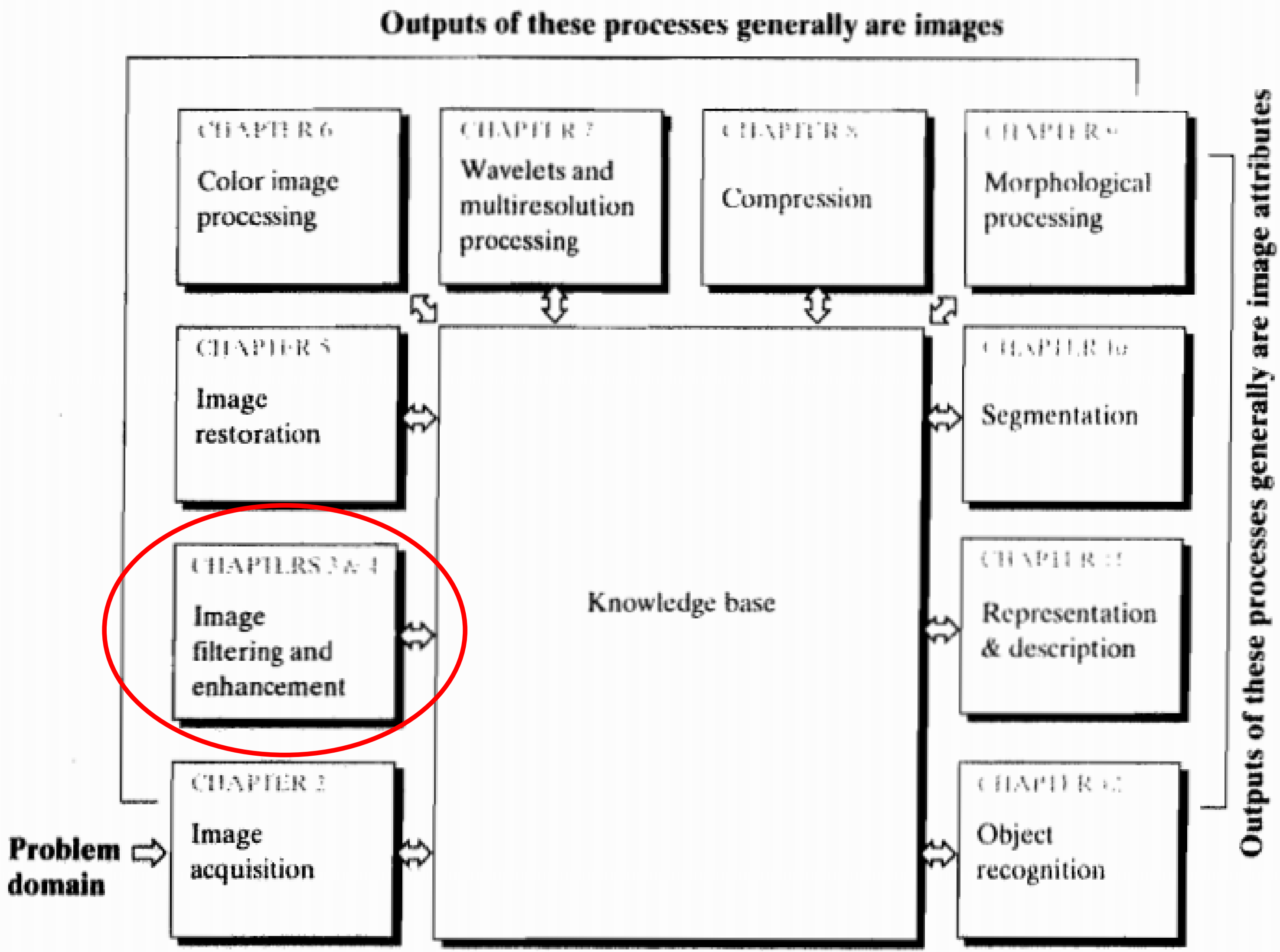
Chapter 10

Segmentation

Remember?

FIGURE 1.23 Fundamental steps in digital image processing. The chapter(s) indicated in the boxes is where the material described in the box is discussed.



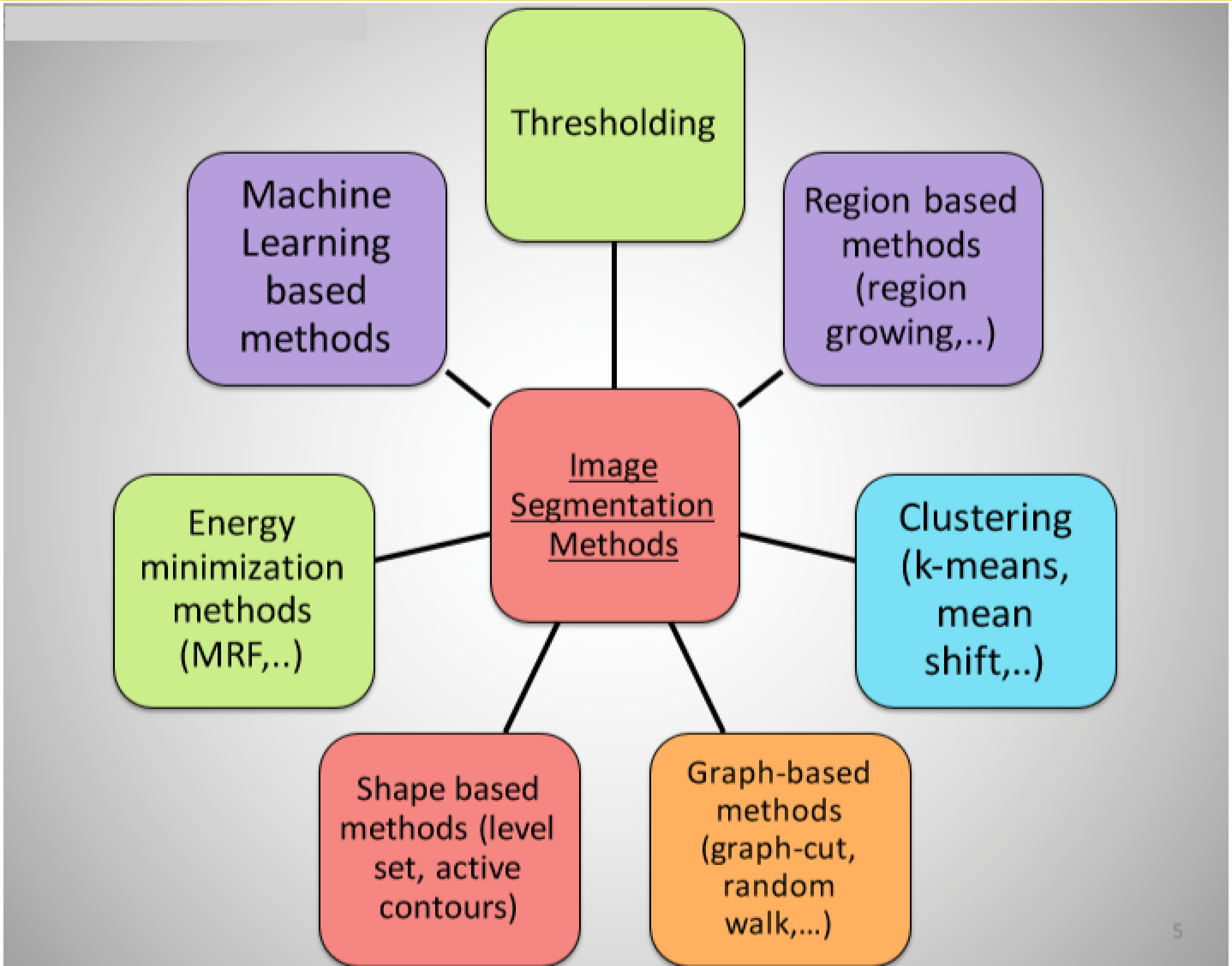
Segmentation !!!

Segmentation

Aim: to partition an image into a collection of set of pixels

- Meaningful regions (coherent objects)
- Linear structures (line, curve, …)
- Shapes (circles, ellipses, …)

Segmentation



Basics of Image Segmentation

Definition: Image segmentation partitions an image into regions called segments.

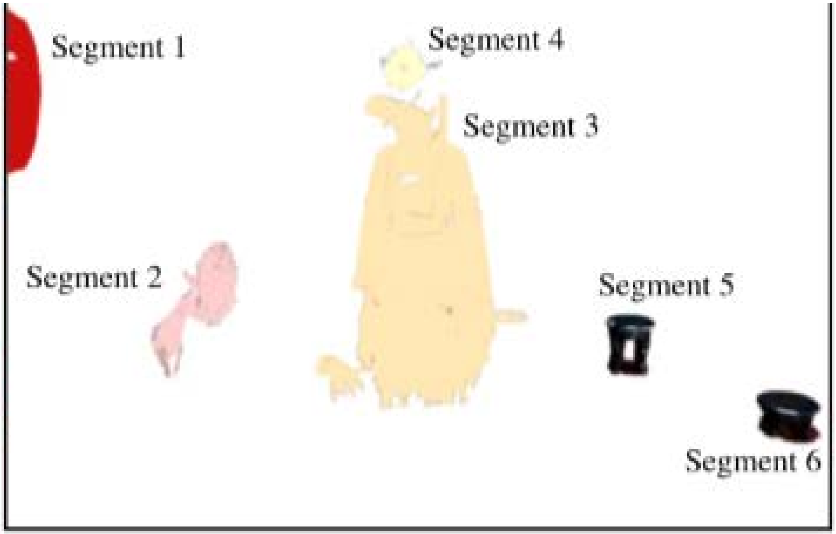


Image segmentation creates segments of connected pixels by analyzing some similarity criteria: intensity, color, texture, histogram, features, …

Basics of Image Segmentation

- Segmentation partitions an image into distinct regions containing each pixels with similar attributes.
- To be meaningful and useful for image analysis and interpretation, the regions should strongly relate to depicted objects or features of interest.

Basics of Image Segmentation

- Meaningful segmentation is the first step from low-level image processing transforming a greyscale or color image into one or more other images to high-level image description in terms of features, objects, and scenes. The success of image analysis depends on reliability of segmentation, but an accurate partitioning of an image is generally a very challenging problem

Image Binarization

• Image binarization applies often just one global threshold T for mapping a scalar image I into a binary image.

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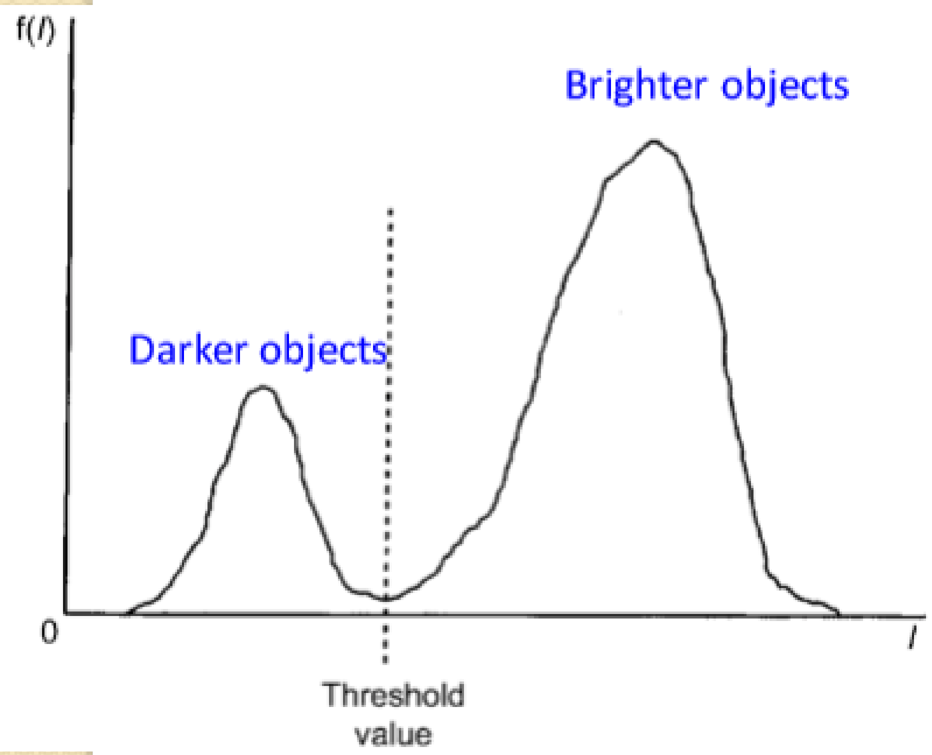
Image binarization applies often just one global threshold T for mapping a scalar image I into a binary image.

• The global threshold can be identified by an optimization strategy aiming at creating “large” connected regions and at reducing the number of small-sized regions, called artifacts

Thresholding

- **Thresholding:** Most frequently employed method for determining threshold is based on histogram analysis of intensity levels



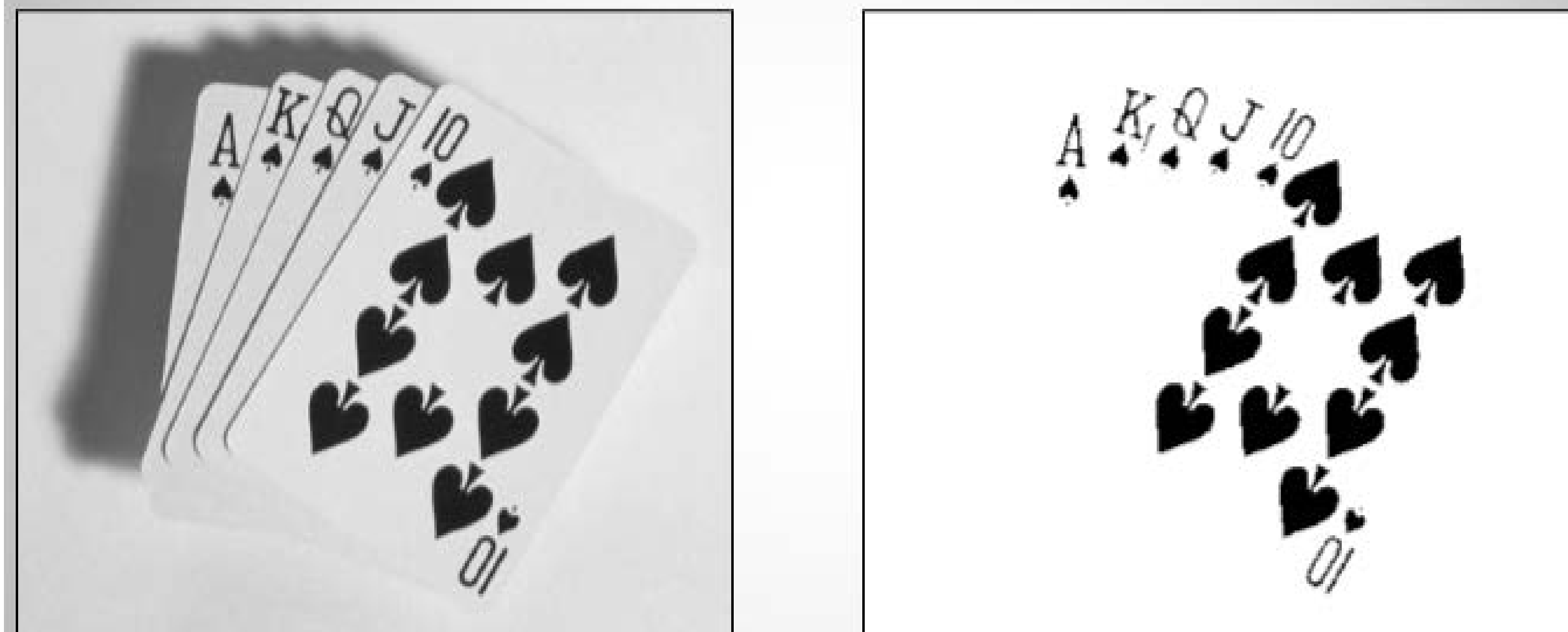
Peak on the left of the histogram corresponds to dark objects

Peak on the right of the histogram corresponds to brighter objects

DIFFICULTIES

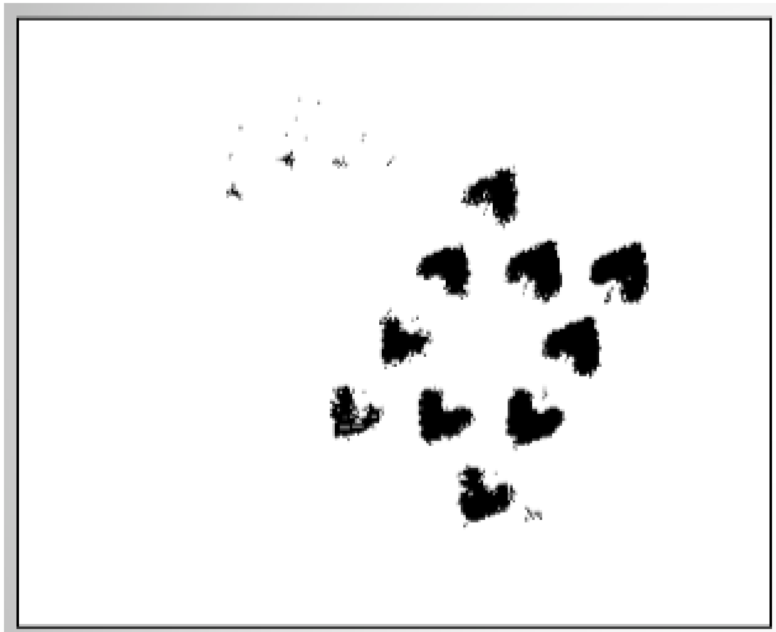
1. The valley may be so broad that it is difficult to locate a significant minimum
2. Number of minima due to type of details in the image
3. Noise
4. No visible valley
5. Histogram may be multi-modal

Thresholding

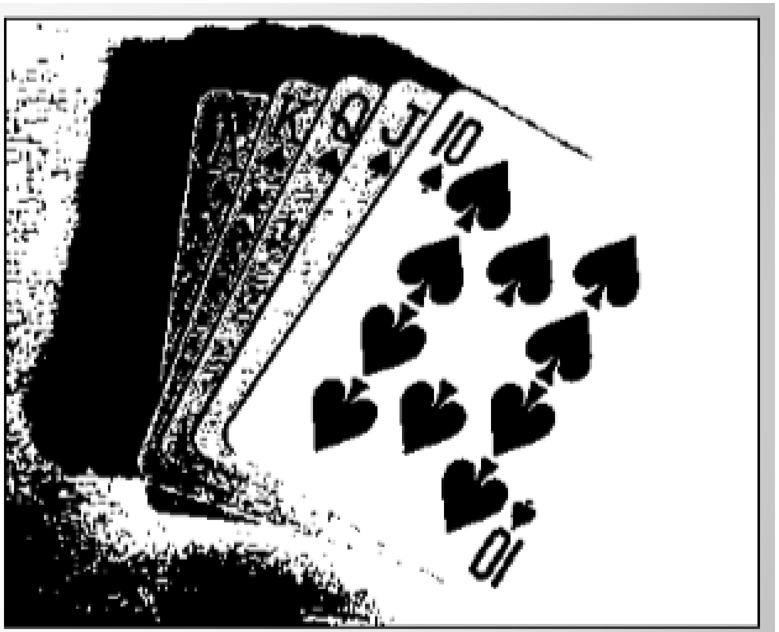


Original Image | Thresholded Image

Thresholding

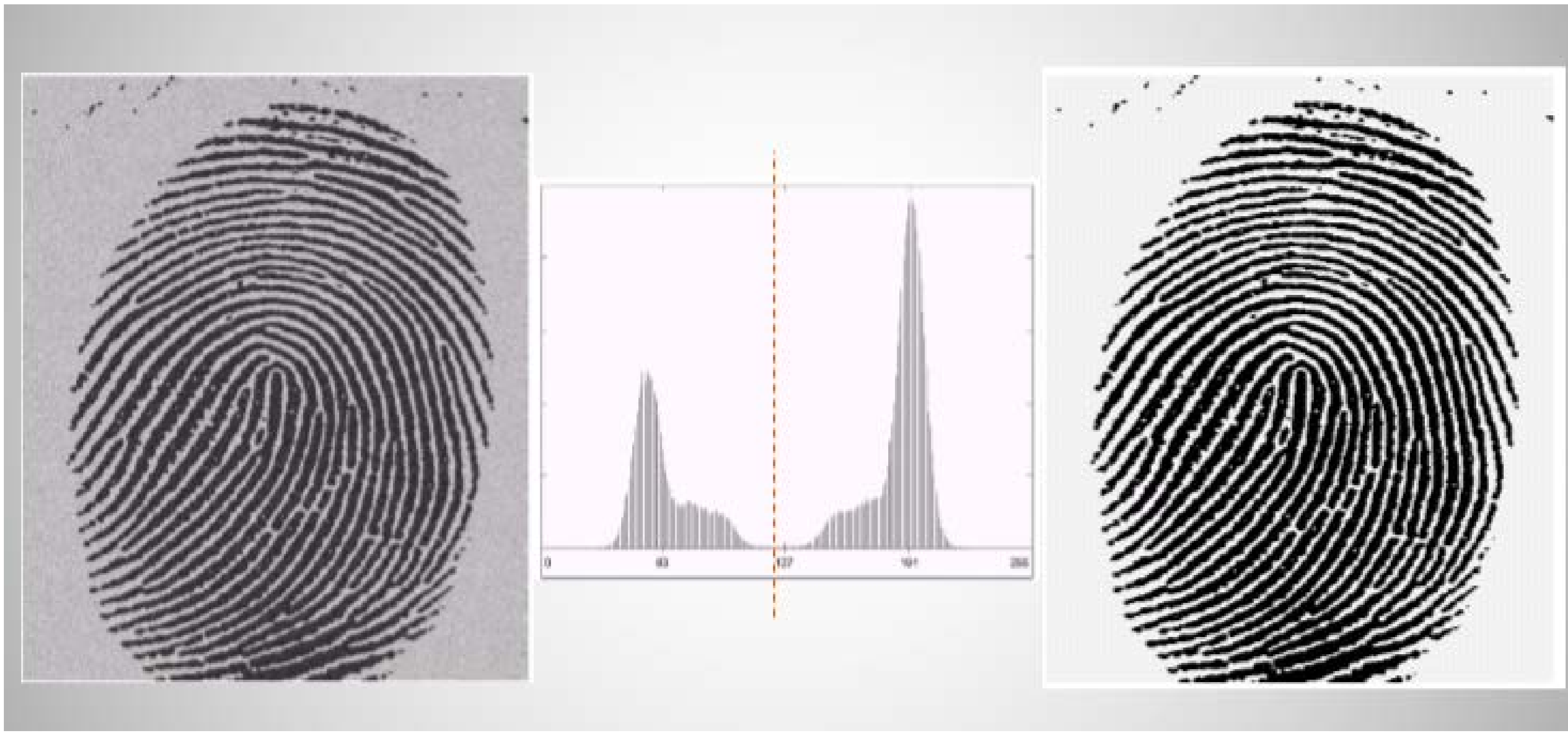


Threshold Too Low

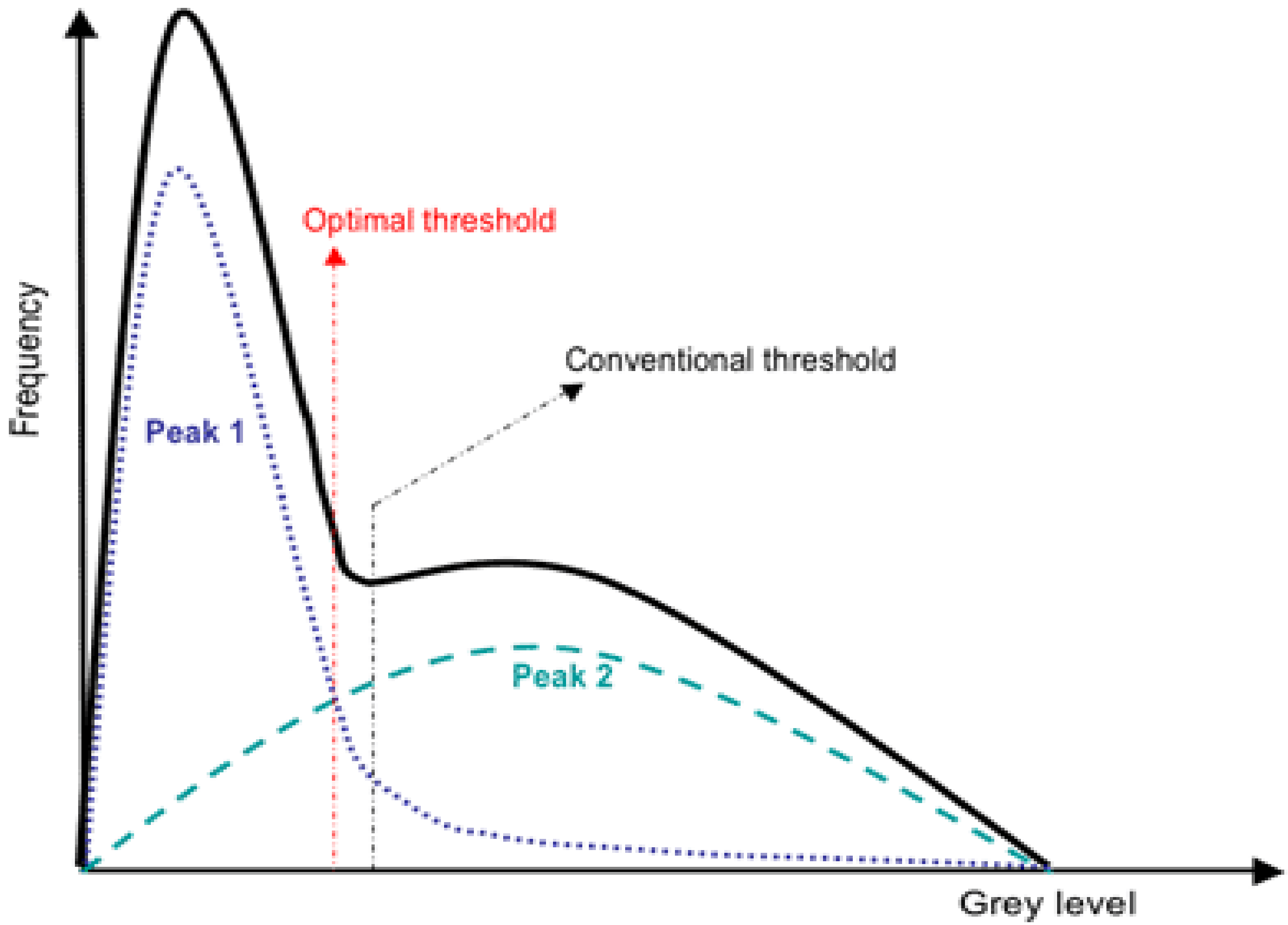


Threshold Too High

Thresholding



Thresholding



OTSU

Definition: The method uses the grey-value histogram of the given image I as input and aims at providing the best threshold in the sense that the "overlap" between two classes, set of object and background pixels, is minimized (i.e., by finding the best balance).

- Otsu's algorithm selects a threshold that maximizes the between-class variance σ_b^2. In the case of two classes,

$$\sigma_b^2 = P_1(\mu_1 - \mu)^2 + P_2(\mu_2 - \mu)^2 = P_1 P_2(\mu_1 - \mu_2)^2$$

where P_1 and P_2 denote class probabilities, and μ_i the means of object and background classes.

- Let c_I be the relative cumulative histogram of an image I, then P_1 and P_2 are approximated by $c_I(u)$ and $1 - c_I(u)$ respectively.
- u is assumed to be the chosen threshold.

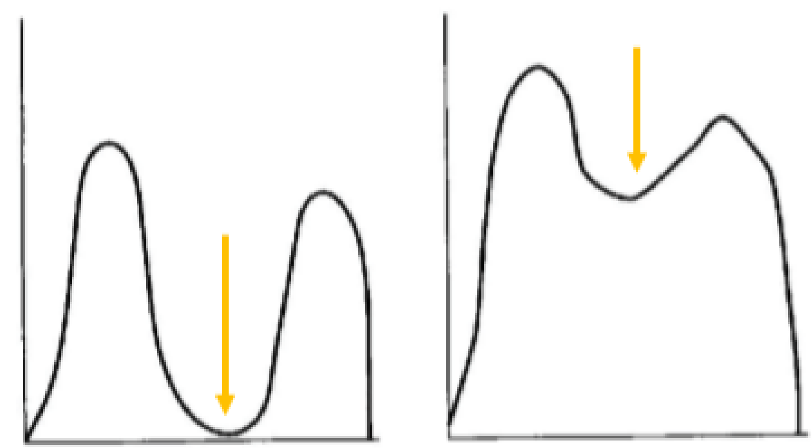
OTSU

Compute histogram H_I for $u = 0, \ldots, G_{\max}$;
Let T_0 be the increment for potential thresholds; $u = T_0$; $T = u$; and $S_{\max} = 0$;
while $u < G_{\max}$ **do**
    Compute $c_I(u)$ and $\mu_i(u)$ for $i = 1, 2$;
    Compute $\sigma_b^2(u) = c_I(u)[1 - c_I(u)][\mu_i(u) - \mu_2(u)]^2$;
    if $\sigma_b^2(u) > S_{\max}$ **then**
        $S_{\max} = \sigma_b^2(u)$ and $T = u$;
    end if
    Set $u = u + T_0$
end while

$$P_1 = \sum_{i=0}^{u} p(i) \qquad \mu_1 = \sum_{i=0}^{u} ip(i)/P_1$$

$$P_2 = \sum_{i=u+1}^{G_{max}} p(i) \qquad \mu_2 = \sum_{i=u+1}^{G_{max}} ip(i)/P_2$$

probabilities   Class means



(a) Two distinct modes   (b) Overlapped modes

Thanks !!!